LLMInsight

Metrics Reference — Definitions, Types & Value Ranges

This document lists every metric available in LLMInsight, grouped by evaluation type. Each entry shows the metric identifier, its expected value type, and a concise description of what it measures and how to interpret it.

GENERIC METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
general_faithfulness	float [0-1]	Checks whether the model output is grounded in and consistent with the provided context or source material.
hallucination	float [0-1]	Identifies factual claims in the output that are not supported by the context or known facts.
answer_relevance	float [0-1]	Measures how directly and completely the response addresses the user's question.
moderation	bool / float	Detects harmful, offensive, or policy-violating content in the output.
completeness	float [0-1]	Evaluates whether all required information or aspects of the question are covered in the answer.

HEURISTIC METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
equals	bool	Exact-match check — passes when output equals the expected reference string.
contains	bool	Substring check — passes when the expected string appears within the output.
is_json	bool	Validates that the output is well-formed, parseable JSON.
matches_regex	bool	Regular-expression check — passes when the output matches the given pattern.
levenshtein	float [0-1]	Computes the edit-distance ratio between output and reference (0 = identical, 1 = completely different).

SUMMARIZATION METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
summary_completeness	float [0-1]	Assesses whether the summary retains all key facts and points from the source.

METRIC NAME	VALUE TYPE	DESCRIPTION
conciseness	float [0-1]	Measures how succinct the summary is relative to the information it conveys.
compression_ratio	float	Ratio of summary length to source length — lower means more aggressive compression.

TRANSLATION METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
comet_score	float [0-1]	Neural MT quality metric (COMET) estimating translation adequacy and fluency on a 0-1 scale.

CODE METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
compilable	bool	Checks whether the generated code compiles / parses without syntax errors.
code_relevance	float [0-1]	Evaluates how well the generated code addresses the given programming task.

RAG METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
context_precision	float [0-1]	Fraction of retrieved context chunks that are actually relevant to the query.
context_recall	float [0-1]	Fraction of relevant information that was successfully retrieved from the context.
contextual_faithfulness	float [0-1]	Verifies the answer is grounded exclusively in the retrieved context with no external fabrication.

AGENT SPAN METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
is_json	bool	Validates that the output is well-formed, parseable JSON.
output_format_compliance	bool / str	Checks that the span output conforms to the expected schema or format specification.
token_efficiency	float [0-1]	Ratio of useful-information tokens to total tokens consumed in a single span.

METRIC NAME	VALUE TYPE	DESCRIPTION
latency_classification	str (Fast/Normal/Slow)	Classifies span latency (Fast / Normal / Slow) against configurable thresholds.

AGENT TRACE METRICS

METRIC NAME	VALUE TYPE	DESCRIPTION
task_completion	float [0-1]	Binary or graded assessment of whether the agent accomplished its overall goal.
step_efficiency	float [0-1]	Ratio of minimal required steps to actual steps taken — penalises redundancy.
tool_error_rate	float [0-1]	Proportion of tool calls across the trace that returned an error.
loop_detection	bool	Flags repetitive cycles of identical or near-identical steps in the trace.
cost_efficiency	float	Total task value delivered per unit cost (tokens × price) across the trace.
action_consistency	float [0-1]	Measures coherence between the agent's stated plan and its actual tool/action choices.
total_cost_summary	dict	Aggregated token usage and estimated USD cost across all spans in the trace.